

Project Assignment 2

EQ2401 Adaptive Signal Processing

Oguzhan Oguz

Personnummer: 20020308-6293 — Email: oguzhano@kth.se

KTH Royal Institute of Technology

Problem Formulation

Objective: Reduce a periodic tonal disturbance corrupting a speech signal using adaptive filtering.

Signal model:

$$y(n) = s(n) + v(n)$$

- $s(n)$: speech signal
- $v(n)$: tonal disturbance

Approach: Adaptive Line Enhancer (ALE) with delay Δ , and comparison of:

LMS, NLMS, RLS

Assumptions and Approximations

Model assumptions

- **Additive model:** $y(n) = s(n) + v(n)$
- **Delay-based correlation separation:**

$$|E[s(n)s(n - \Delta)]| \ll |E[v(n)v(n - \Delta)]|$$

for a suitable Δ

- **Weak cross-correlation assumption:**

$$E[s(n)v(n - k)] \approx 0$$

Approximations in implementation

- Tonal component is modeled by a finite-length FIR filter ($M = 128$)
- LMS / NLMS use instantaneous error and instantaneous energy instead of exact expectations
- RLS uses exponential forgetting ($\lambda < 1$) to model slowly varying statistics
- LMS step-size uses a conservative trace-based bound instead of exact $\lambda_{\max}$

Adaptive Line Enhancer (ALE) Architecture

ALE signal definitions

- Delayed reference: $u(n) = y(n - \Delta)$
- Target: $d(n) = y(n)$
- Filter output (tone estimate): $\hat{v}(n) = \mathbf{w}^T(n-1)\mathbf{u}(n)$
- Error - enhanced speech estimate: $e(n) = d(n) - \hat{v}(n)$

Why the delay Δ helps

- Speech is broadband, so correlation decreases as delay increases
- Tonal disturbance is periodic, so it remains predictable after delay
- Therefore, the adaptive filter mainly learns the tonal disturbance, and the error signal approaches the cleaner speech component

Least Mean Squares (LMS)

Instantaneous cost

$$J(n) = e^2(n) = (d(n) - \mathbf{w}^T(n-1)\mathbf{u}(n))^2$$

Update rule

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mu \mathbf{u}(n) e(n)$$

Step-size bound

$$0 < \mu < \frac{2}{\lambda_{\max}}$$

$$\lambda_{\max} \leq \text{tr}(\mathbf{R}_u) = \sum_i \lambda_i = E[\|\mathbf{u}(n)\|^2] \approx MP_u$$

$$\Rightarrow \mu_{\max}^{(\text{cons})} = \frac{2}{MP_u} \leq \frac{2}{\lambda_{\max}}$$

- M : filter length, P_u : average power of delayed reference
- In code: $\mu_{\text{LMS}} = 0.05 \mu_{\max}^{(\text{cons})}$

Normalized LMS (NLMS)

Motivation: Normalize the adaptation step using the instantaneous input energy.

Update rule

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \frac{\mu_{\text{NLMS}}}{\epsilon + \mathbf{u}^T(n)\mathbf{u}(n)} \mathbf{u}(n) e(n)$$

Key parameters

- $\mathbf{u}^T(n)\mathbf{u}(n)$: instantaneous energy of the input buffer
- ϵ : small regularization constant
- $0 < \mu_{\text{NLMS}} < 2$: normalized step-size factor

Expected behavior: More robust than LMS under non-stationary speech amplitude.

Recursive Least Squares (RLS)

Exponentially weighted least-squares cost

$$J(n) = \sum_{i=1}^n \lambda^{n-i} e^2(i), \quad 0 < \lambda \leq 1$$

Recursive updates

$$\mathbf{k}(n) = \frac{\mathbf{P}(n-1)\mathbf{u}(n)}{\lambda + \mathbf{u}^T(n)\mathbf{P}(n-1)\mathbf{u}(n)}$$

$$e(n) = d(n) - \mathbf{w}^T(n-1)\mathbf{u}(n)$$

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{k}(n)e(n)$$

$$\mathbf{P}(n) = \frac{1}{\lambda} [\mathbf{P}(n-1) - \mathbf{k}(n)\mathbf{u}^T(n)\mathbf{P}(n-1)]$$

Initialization

$$\mathbf{P}(0) = 10000 \mathbf{I}$$

High initial uncertainty gives fast initial adaptation.

Chosen Parameters and Design Choices

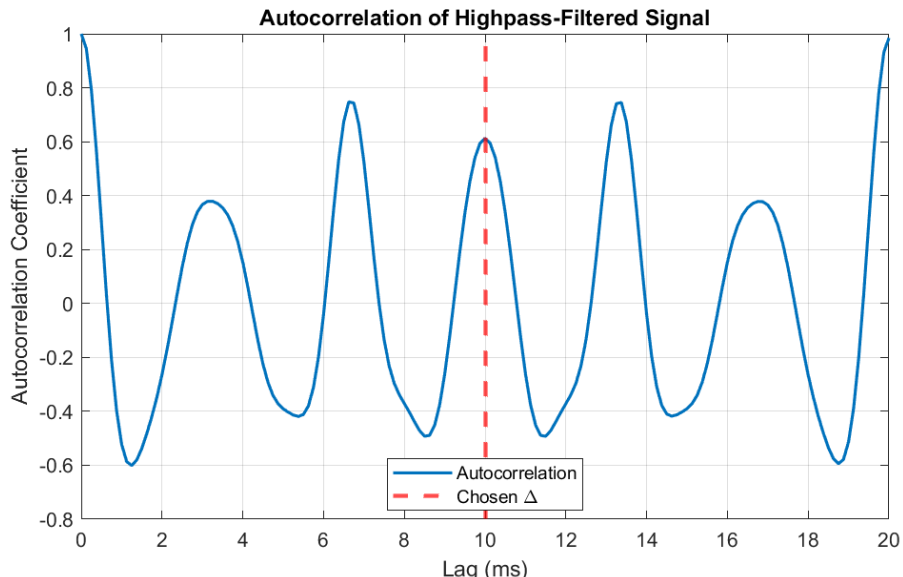
Delay selection

- Selected delay: $\Delta = 80$ samples = 10.00 ms
- Chosen from the autocorrelation trend of a highpass-filtered signal
- Trade-off:
 - large enough to reduce speech correlation
 - small enough to preserve tonal correlation

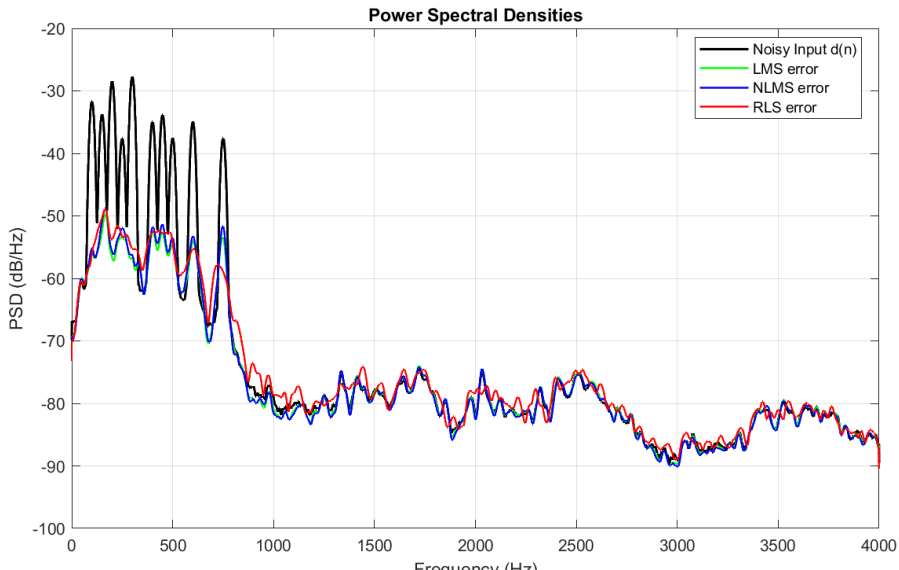
Adaptive parameters

- LMS: $\mu_{\text{LMS}} = 0.05 \mu_{\text{max}}^{(\text{cons})}$, $\mu_{\text{max}}^{(\text{cons})} = \frac{2}{MP_u}$
- NLMS: $\mu_{\text{NLMS}} = 0.05$, $\epsilon = 10^{-6}$
- RLS: $\lambda = 0.9995$, $\mathbf{P}(0) = 10000 \mathbf{I}$

Delay Motivation: Autocorrelation of Highpass-Filtered Signal



Results: Power Spectral Densities



Evaluation Methodology and Tone-Reduction Metric

Evaluation assumptions / approximations

- PSD is estimated from finite data (Welch estimate)
- f_0 is approximated by the largest PSD peak in a selected search range
- Harmonic power is approximated by summing spectral bins around kf_0

Tone-band power and reduction

$$T = 10 \log_{10} \left(\sum_{f_i \in \mathcal{B}} P(f_i) \Delta f \right)$$

$$\text{Tone reduction (dB)} = T_{\text{in}} - T_{\text{out}}$$

Quantitative Results and Conclusion

Measured harmonic tone-reduction results

Method	Tone-band power (dB)	Tone reduction (dB)
Input	$T_{\text{in}} = -18.72$	–
LMS	$T_{\text{LMS}} = -43.94$	25.22
NLMS	$T_{\text{NLMS}} = -43.16$	24.44
RLS	$T_{\text{RLS}} = -43.84$	25.12

Main observations

- All methods provide strong tonal suppression.
- LMS and RLS are very close in this run and give the highest reductions.
- NLMS is slightly lower in this metric, but remains robust and stable.
- Complexity trade-off: $\text{LMS} < \text{NLMS} < \text{RLS}$.